



Narrator identification by querying Sanad graph and utilizing the NarratorsKG on AR-Sanad 280K-v2 dataset

Somaia Mahmoud¹ · Emad Nabil² · Omar Saif³ · Marwan Torki¹

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Abstract

Narrator disambiguation is a field within hadith science that studies unidentified narrators in hadith narration chains, also known as sanads. Sanads can be represented as graphs, with the nodes representing the narrators and the edges representing their relationships in the chain. The current methods for resolving the narrator disambiguation problem do not utilize the graph structure of the sanad, but by leveraging this structure, we can apply graph computational and deep learning techniques to identify narrators. This paper introduces a method that utilizes the sanad graph structure to identify all narrators in a given sanad. Our two-stage approach begins by generating a query embedding and identifying the top k narrator entities closest to the query embedding. We then use AraBERT to re-rank the top k narrators and make the final prediction. Our method achieves 94.6% accuracy on the validation set of AR-Sanad 280K dataset. Additionally, we present AR-Sanad 280K-v2, an updated dataset that represents real hadiths more accurately.

Keywords Knowledge graph · Graph query embedding · Hadith · Narrator identification · Sanad dataset

1 Introduction

In Islam, the term “Hadith” refers to the report of the sayings and actions of Prophet Muhammad (PBUH). A Hadith comprises of two essential parts: the “matn,” which is the content of the narration, either the words or the deeds of the Prophet, and the “sanad,” which is the chain of narrators who conveyed the Hadith from the era of Prophet Muhammad (PBUH) to the present time. An example of a Hadith is displayed in Fig. 1, where the sanad is underlined. Hadiths

are integral to the Qur’an as they help to explain and illustrate the teachings of the Qur’an. Muslims rely on Hadiths to determine what is permissible or prohibited by the laws of Islam, hence it is crucial to ensure that Hadiths are not fabricated.

The science of hadith has various disciplines that deal with the documentation and authentication of hadith. The laws of hadith cover all possibilities of strength or weakness in both the matn and the sanad. Hadiths are categorized by scholars based on their authenticity, with levels ranging from the most authentic (أصح الأسانيد) to counterfeit and fabricated (الموضوع والمذلس). Other rankings include fair by itself (الحسن لذاته) and fair by external consideration (الحسن لغيره), as well as weak and discarded hadiths [1] (المتروك والضعيف).

To verify the authenticity of hadiths, scholars use five criteria: (1) the sanad cohesion, (2) the morality of the narrators, (3) the accuracy of their narratives, (4) anomalies, (5) and flaws in the sanad and matn [2]. To find information about narrators’ morality and the credibility of their narratives, scholars refer to books about narrators’ invalidation and rectification “كتب الجرح والتعديل” including what early scholars wrote about them. And to identify anomalies and flaws, they compare the investigated hadith with similar narrations of other reliable narrators.

✉ Emad Nabil
emadnabil@iu.edu.sa

Somaia Mahmoud
somaia.0246@gmail.com

Omar Saif
osaif@iu.edu.sa

Marwan Torki
mtorki@alexu.edu.eg

¹ Department of Computer and Systems Engineering,
Alexandria University, Alexandria, Egypt

² Faculty of Computer and Information Systems, Islamic
University of Madinah, Madinah 42351, Saudi Arabia

³ Faculty of Hadith and Islamic Studies, Islamic University
of Madinah, Madinah, Saudi Arabia

Fig. 1 An example hadith from Sahih Al-Bukhari book. The sanad is underlined and narrator names are in blue. Best seen in color

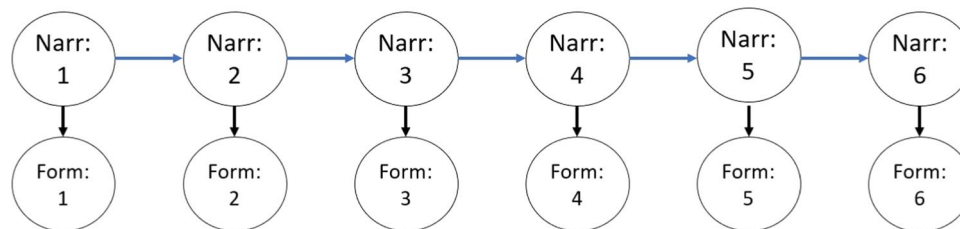
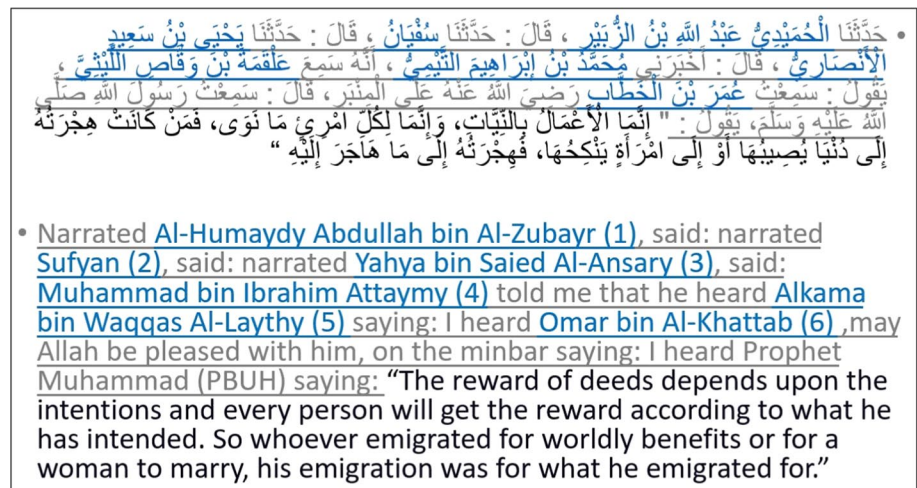


Fig. 2 Graph representation of the sanad in Fig. 1, showing nodes Narr:i and Form:i for each narrator and appearance form, respectively. The black arrows indicate the relation between narrators and

their appearance forms, while the blue arrows indicate the “narrated from” relation. Best seen in color

In order to obtain information about the narrators mentioned in the investigated sanad, hadith scholars need to first identify the narrators themselves. This process, known as narrator disambiguation, is a complex process and it holds significant importance. Misidentifying the narrators can lead to incorrect judgments regarding the authenticity of the hadith. Therefore, it is of utmost importance to pay great attention to this process [1].

Ambiguity arises as a result of shared appearance forms among multiple narrators. The phrase “appearance form” refers to the way that narrator names appeared in the sanad, as per [3]. In Fig. 1, the blue names indicate the appearance forms. For instance, the name Sufyan in Fig. 1 exemplifies the issue of ambiguous names. In other words, the name Sufyan could refer to Sufyan bin Uyayna or Sufyan Al-Thawry due to the shared appearance form. In addition to first names, there are also common bynames and teknonyms that are shared among multiple narrators.

This paper focuses on solving this issue by leveraging the graph structure of the sanad. Sanads have a distinctive nature that enables their representation through graphs, which is not feasible for other textual data. The graph in Fig. 2 demonstrates the representation of the sanad illustrated in Fig. 1. We used nodes to represent the narrators

(Narr:i) and their appearance forms (Form:i), while the edges indicate the narration flow between narrators (represented by blue arrows) and the relation between narrators and their appearance forms (represented by black arrows). The “narrated from” relationship is denoted by the blue arrows and can be reversed to denote the “narrated to” relationship. We distinguish between narrator nodes and appearance form nodes as the appearance forms are not unique to a single narrator.

For our experiments, we use the AR-Sanad 280K dataset [3].¹ It contains a training set of synthetic sanads and a test set of real hadith sanads for a realistic evaluation of models. In order to improve the representativeness of the synthetic training set, we made some adjustments. In the AR-Sanad 280K dataset, sanads were generated by randomly selecting appearance forms for the narrators. However, in AR-Sanad 280K-v2 dataset, we sample appearance forms from the distribution of the appearance forms of each narrator. As a result, training on the AR-Sanad 280K-v2 dataset yielded significantly better results on the real hadiths test set.

¹ <https://github.com/somaia02/Narrator-Disambiguation>.

This paper presents the first knowledge graph of hadith narrators, called NarratorsKG. This graph represents narrators and their relationships. We also propose a new method that utilizes the sanad graph structure to identify narrators in hadith sanads by linking appearance forms in the sanad to narrator entities in the graph.

Our model aims to assist hadith scholars in their work, but it is important to note that the final decision about narrator identities should not rely solely on the predictions made by the model. To summarize, this paper makes several contributions:

- **Model:** A novel model is presented that identifies narrators in the sanad by exploiting the graph structure of the sanad.
- **AR-Sanad-v2 dataset:** The training set from the AR-Sanad 280K dataset is updated by sampling from the distribution of appearance forms for each narrator rather than uniform sampling in the AR-Sanad 280K dataset.
- **Narrators' Graph:** NarratorsKG is a graph of narrators, their interactions, and their appearance forms.

2 Related work

2.1 Hadith

In recent years, there has been remarkable advancement in utilizing software solutions to address problems related to hadith science. Several surveys provide an overview of most of the works in this field [4, 5]. Some works have been done to visualize the narration graph for hadiths with single or multiple narration chains using hadith sanads [6–9]. Other works have focused on creating a network of narrators and carrying out social network analysis [10–15].

Ramzy et al. [16] created a dataset of 29K hadiths and classified them into 3 categories based on the author of the hadith. The author could be Prophet Muhammad, his companions, or their successors. Our work only deals with hadiths of Prophet Muhammad. Our aim is to identify the chain of narrators who heard the hadith from Prophet Muhammad to the present day. This is a different problem from author identification.

Astari et al. [17] used word2vec and cosine similarity to determine if the same appearance form (mention) in 2 different sanads refers to the same narrator. However, they do not link the appearance forms to narrator entities. Prasetyo et al. [18] used a KNN classifier to connect narrator mentions in sanads to narrator entities in DBpedia. However, DBpedia contains very few narrator entities, and only one match was found among the six examples presented. Mahmoud et al. [3] fine-tuned AraBERT to identify narrators. They

create a list of 18K narrators and link mentions of narrators in sanads to narrator entities from the list.

2.2 Graph-based entity resolution

The field of entity resolution involves identifying and resolving references to the same real-world entity across different data sources, which can be challenging due to variations in entity representations, inconsistencies, and data errors. Graph-based approaches have gained significant attention in entity resolution tasks due to their ability to capture complex relationships and dependencies between entities. This makes them highly applicable to the problem of narrator disambiguation. This section gives a brief overview of the most relevant works.

Varma et al. [19] presented a method to resolve ambiguity in biomedical entities, which involves three steps: candidate extraction, reranking, and post-processing. They train BERT [20] to extract candidates and use the trained weights to initialize the reranker. The idea of the reranker is similar to GlossBERT [21] which utilizes definitions of word senses to disambiguate words. In the reranker, entities' metadata are used in a similar manner to the sense definitions.

Hamilton et al. [22] proposed a method for answering conjunctive logical queries over incomplete graphs that may have missing edges. They embed queries and entities into a low-dimension space so that entities corresponding to the answer of a query are embedded close to the query. Other approaches use similar methods to handle more complex queries, such as embedding queries as boxes [23] or probabilistic distribution [24]. Another work uses RoBERTa [25] to embed natural language queries [26].

3 Resources

This section introduces the resources used in this research. We run our experiments on the AR-Sanad 280K dataset. We also describe how we updated the dataset to be more similar to real sanads. The last subsection describes how we constructed our knowledge graph *NarratorsKG*.

3.1 AR-Sanad 280K dataset

The AR-Sanad 280K dataset consists of a training set of 279,625 artificially generated sanads, where the narrators are identified, and a test set of 27,056 real sanads from the six most famous hadith books. The process of generating the artificial training sanads involves initially creating a narration chain that consists of narrator IDs. Subsequently, for each narrator, an appearance form is randomly chosen from their available appearance forms.

Fig. 3 Comparison of two sanads from the AR-Sanad 280K dataset and the updated version AR-Sanad 280K-v2 dataset. For the same narrator IDs list, the appearance forms selected in the v2 dataset are the most frequently used forms

• Narrator IDs: [400,1171,120] • V1 sanad: ابن كيسان [فاصل] حميد [فاصل] رجل من أصحاب النبي صلى الله عليه وسلم قال أظنه ابن عمر • V2 sanad: صالح بن كيسان [فاصل] حميد [فاصل] ابن عمر
• Narrator IDs: [88,11197,1998,559] • V1 sanad: الحجاج يعني ابن أرمطة [فاصل] عبد الله بن عبد الرحمن [فاصل] عبد الرحمن يعني ابن أبي ليلي [فاصل] أبو بكر • V2 sanad: حجاج [فاصل] عبد الله بن عبد الله [فاصل] عبد الرحمن بن أبي ليلي [فاصل] أبو بكر الصديق
• Narrator IDs: [400,1171,120] • V1 sanad: Bin Kaysan [SEP] Humayd [SEP] a companion of Prophet Muhammad (PBUH) thought to be bin Omar • V2 sanad: Saleh bin Kaysan [SEP] Humayd [SEP] bin Omar
• Narrator IDs: [88,11197,1998,559] • V1 sanad: Al-Hajjaj i.e. bin Artoah [SEP] Abdullah bin Abdulrahman [SEP] Abdulrahman i.e. bin Abi Layla [SEP] Abu Bakr • V2 sanad: Hajjaj [SEP] Abdullah bin Abdullah [SEP] Abdulrahman bin Abi Layla [SEP] Abu Bakr Al-seddiq

The choice to create artificial sanads for training is driven by the scarcity of a large digital collection of real hadiths suitable for training purposes. Nonetheless, despite their artificial nature, these training sanads can still provide valuable insights for disambiguating names within real sanads. This is due to the fact that any two narrators are chained together only if they have narrated from each other in an actual hadith.

3.2 AR-Sanad 280K-v2

We update the training and validation sets of the AR-Sanad 280K dataset, while leaving the test set unchanged as it comprises of real sanads. The chains of narrators (IDs list) in the training and validation sets are not altered. The narrators are the same only their appearance forms are modified. Rather than randomly selecting appearance forms during the generation of the training set, we select them from a probability distribution that is dependent on how frequently the narrator is referred to by the specific appearance form.

Figure 3 illustrates two ID lists. For each ID list, we show two sanads that have been generated—one from the original AR-Sanad 280K dataset, and the other from the updated version of the dataset. In the first sanad, narrator 120 (Abdullah bin Omar) is referred to using the appearance form “a companion of Prophet Muhammad (PBUH) thought to be bin Omar,” which was found in only one real hadith. In contrast, in the second sanad, narrator 120 is referred to using the appearance form “bin Omar,” which is the most commonly used form to refer to Abdullah bin Omar and has been found in at least 15,649 real hadiths. We observe the same thing

with the second ID list, all generated appearance forms in the updated dataset are the most used forms except for the last narrator Abu Bakr, and the second most common form is selected. On the contrary, the forms generated in the original dataset are all rare forms except for the last narrator.

The distribution of appearance forms used for narrator Abdullah bin Omar in real hadiths is presented in Fig. 4a. This distribution serves as a basis for sampling appearance forms for this narrator in the AR-Sanad 280K-v2 dataset, as shown in Fig. 4c. In contrast, the distribution in the AR-Sanad 280K dataset is almost uniform, with each appearance form appearing in 150–200 sanads, as illustrated in Fig. 4b.

To identify the narrators, we transform each sanad into a graph query. For a 3-narrator sanad, we create three separate queries, one for each narrator. Figure 5 shows two graph queries for the sanad “Nafie from bin Omar.” We transform all sanads in the dataset, producing 1,139,167 training queries, 284,181 validation queries, and 150,904 test queries.

3.3 Knowledge graph: *NarratorsKG*

The *NarratorsKG* graph is constructed as a directed graph $G = (V, E)$, which consists of nodes $v \in V$ and edges $e \in E$. There are two types of nodes: one type represents narrators and the other represents appearance forms. A partial view of the *NarratorsKG* is depicted in Fig. 6. *NarratorsKG* comprises a total of 79,256 nodes, which include 18,298 narrator nodes and 60,958 appearance form nodes. Additionally, there are 70,415 edges linking the narrators to their respective appearance forms, and 126,768 edges linking the narrators who have narrated from one another.

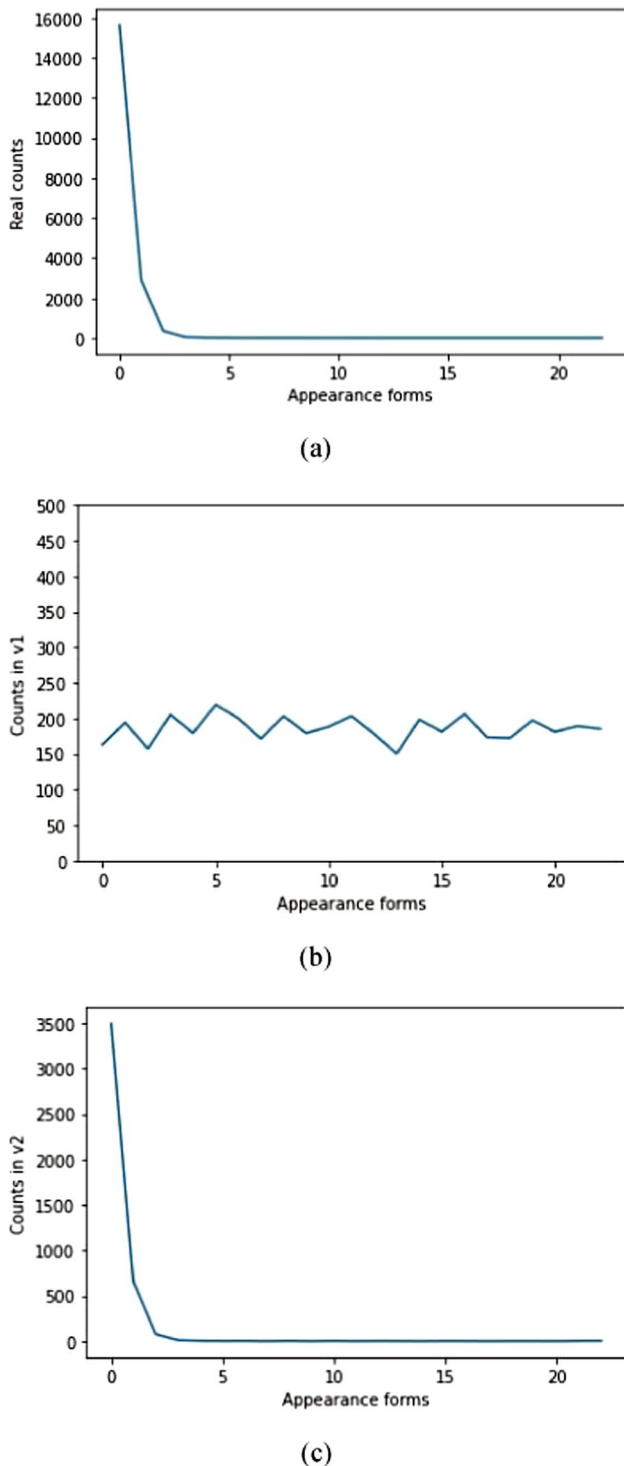


Fig. 4 Distribution of narrator Abdullah bin Omar's appearance forms. **a** The distribution in real hadiths. **b** and **c** are the distribution in 4289 sanads from the AR-Sanad 280K dataset and the AR-Sanad 280K-v2, respectively. The X-axis indicates appearance forms. The Y-axis indicates the number of sanads where they appear referring to Abdullah Bin Omar

Additional information could be incorporated into *NarratorsKG*, such as data on death years, teknonyms, and residence cities, to enrich its content. However, these data were not included as it is in natural language and require processing to extract the relevant entities. Nonetheless, we utilize this information in the second stage of our model. All information about the narrators' (including their interactions, appearance forms, death years, etc.) was obtained from the publicly available data provided by Mahmoud et al. [3].

4 Model

Using a two-stage approach is a common practice in Named Entity Disambiguation (NED) [19]. The initial stage is the candidate generator which extracts a set of plausible candidates. The second stage is the reranker which assigns a score to each candidate. Our model adopts this approach as depicted in Fig. 7. We utilize an embedding-based approach to generate candidates, which is more efficient computationally than using transformer-based models for both the generator and the reranker.

4.1 Generator

Assuming we need to identify the narrators involved in a simple 2-narrator sanad such as "Nafie from bin Omar," we can query the narrators' knowledge graph to obtain their identities, as demonstrated in Fig. 5. However, our knowledge graph includes approximately 80,000 nodes, representing narrators and appearance forms. Even if we prune the search space utilizing information from the sanad, the search space would still be huge. Consequently, as the sanad becomes longer, the time required to obtain results becomes prohibitively long. Moreover, certain queries may result in multiple potential matches, making it challenging to determine a single, definitive identification. To address these issues, we utilize an embedding-based approach that has a time complexity linear in the number of edges in the query and constant relative to the size of the input graph [22]. This approach is more efficient compared to traditional graph searching algorithms, which exhibit complexity proportional to the size of the input graph.

Similar to the approach taken by Hamilton et al. [22], we want to generate query embedding and node embedding so that the query and its answer node are embedded close to each other. We use two operators to obtain a query embedding, the edge operator and the node operator. However, we make modifications to these operators to be more akin to the ones described by Ren et al. [23]. Comparison between the performances of different operators is discussed in Sect. 5.1.

Fig. 5 Queries to identify narrators in the sanad: “Nafie from bin Omar” and graph patterns corresponding to the queries. Best seen in color

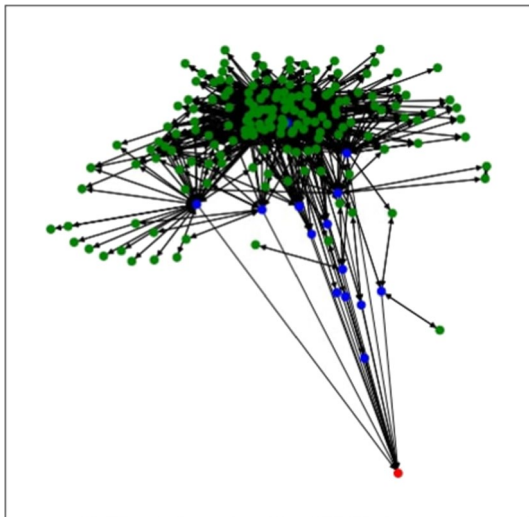
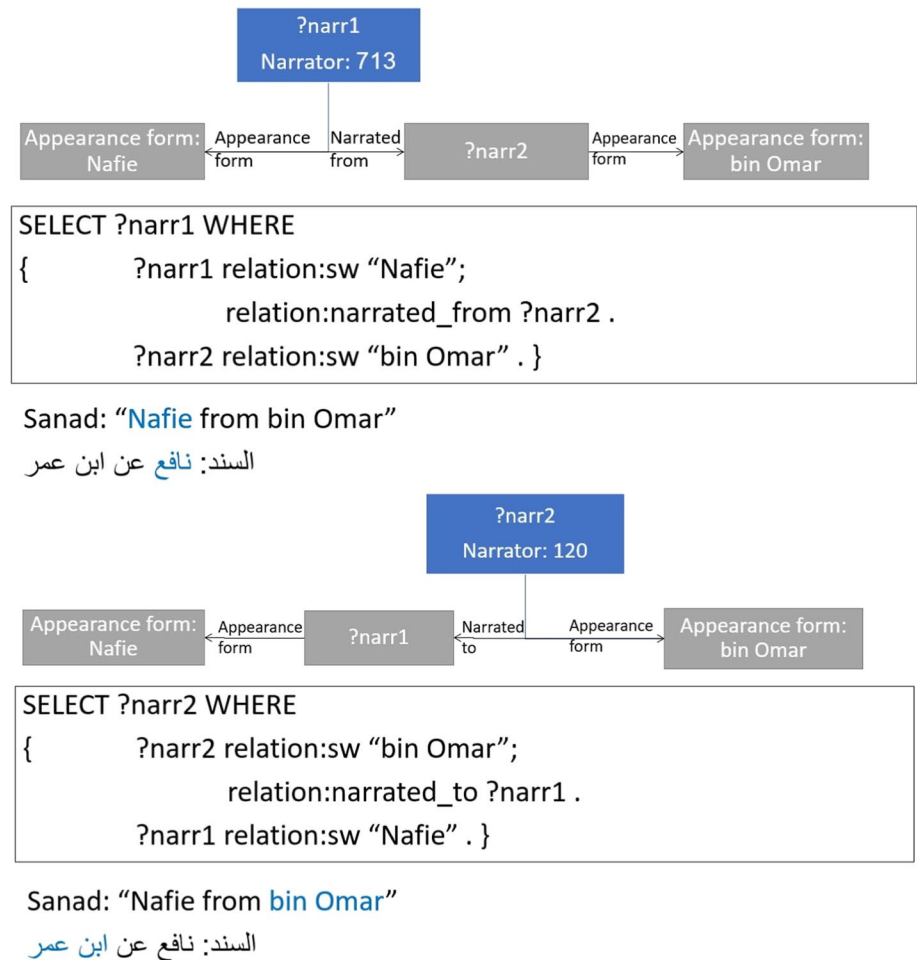


Fig. 6 A part of the *NarratorsKG* graph. The red node represents the appearance form “Nafie,” connected to 14 blue nodes representing narrators who share this appearance form, and 169 green nodes representing narrators who narrated from them. Best seen in color

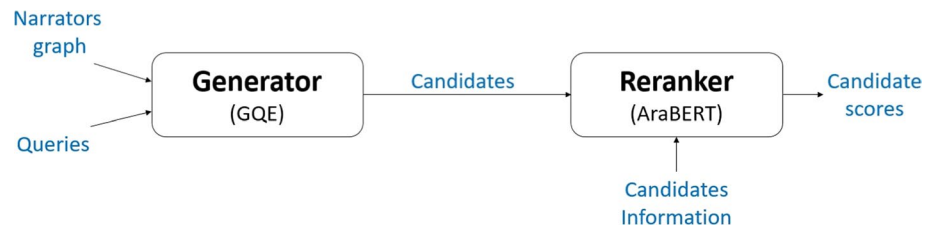
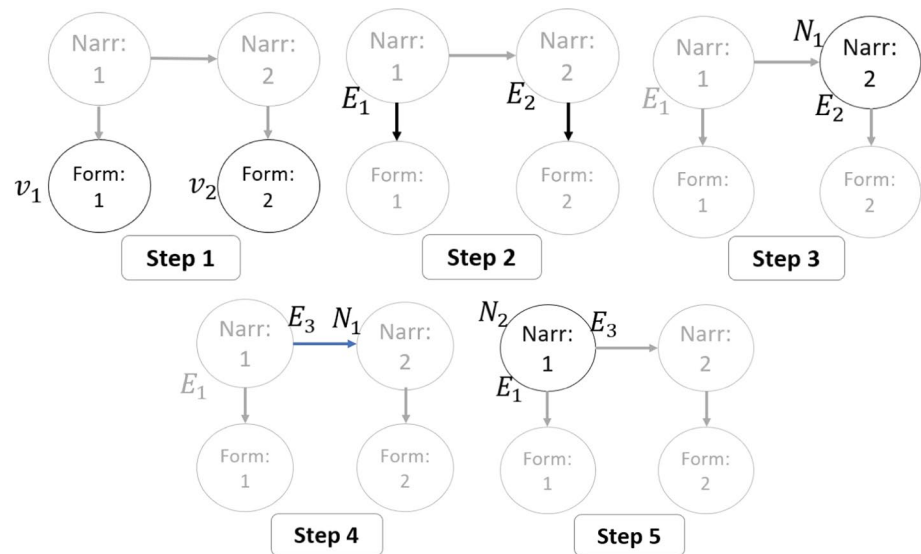
Edge operator: Given two nodes connected by an edge, the operator takes the embedding of the source node and the edge type as input and produces an embedding representing the target node. The nodes and edge types are embedded as vectors v, t . The resulting output can be expressed as:

$$E = v + t \quad (1)$$

Node operator: to obtain a final embedding for a target node, we use the node operator which takes a set of embeddings that represent the target node and outputs a single embedding. Specifically, given n embeddings $E_1, \dots, E_n$, the output can be represented as:

$$N = \sum_i a_i \cdot E_i, \quad a_i = \frac{\exp(E_i)}{\sum_j \exp(E_j)} \quad (2)$$

To obtain the query embedding, we initiate the graph traversal from the source nodes, which are the appearance forms, and apply the two operators iteratively until we reach the target node, which is the identified narrator. The query embedding is the output of the node operator at the target node. We then extract the top k narrators, who share the

Fig. 7 Model illustration**Fig. 8** Generator steps. Step 1: we retrieve v_1, v_2 . Step 2: we compute E_1, E_2 by applying the edge operator. In step 3: $N_1 = E_2$. Step 4: we compute E_3 . Step 5: we compute N_2 by applying the node operator to E_1, E_3 . Best seen in color

same appearance form and whose node embeddings are closest to the query embedding.

Consider the graph pattern depicted in Fig. 5 where we aim to identify the narrator Nafie. The following steps illustrate how we generate a query embedding:

1. We start with the embedding of the two appearance forms, “Nafie” and “bin Omar” (v_1, v_2) Fig. 8 step 1.
2. Then, we apply the edge operator with the relation, appearance form. Then, we get two vectors, E_1, E_2 , one representing each narrator Fig. 8 step 2.
3. At the node (Narr:2), since there is only one outgoing edge, we do not need to apply the node operator. Therefore, N_1 is equal to E_2 Fig. 8 step 3.
4. Then, we apply the edge operator again to N_1 , the vector representing Narr:2, with the relation narrated from. We get E_3 Fig. 8 step 4.
5. Now, we have two vectors representing narrator “Nafie,” E_1, E_3 . We apply the node operator to get the final query embedding, N_2 Fig. 8 step 5.
6. To determine the narrator identity, we compare the query embedding with all nodes representing narrators who share the appearance form “Nafie” and select the top k nodes with the closest embeddings using cosine similarity

4.1.1 Training details

The batch size, learning rate, and embedding size are 128, 0.001, and 512, respectively. We apply early stopping regularization. We use Adam optimizer. Training the generator takes a few hours on a single GPU. The number of parameters in the generator is 40,606,208 which take about 160 MB in memory.

4.2 Reranker

The reranker is designed to assign a score to each candidate based on a given sanad, a target narrator, a set of candidates, and their information. Drawing inspiration from GlossBERT [21], we use AraBERT [27] as a reranker, which takes a context-info pair as input. The context refers to the sanad that contains the narrator to be identified, while the info part contains information about one of the generated candidates. Figure 9 and Table 1 illustrate sample inputs for the reranker. The candidate information includes details such as the narrator’s full name, byname, year of death, country of death, country of residence, visited countries, rank, relatives, and lineage.

Fig. 9 Example of the reranker's input and its English translation. The target narrator is highlighted in blue. The tokens [START] and [END] are used to indicate the target narrator. The information following the [DESC] token pertains to one of the candidates. Best seen in color

• مالك [بداية] نافع [نهاية] ابن عمر [وصف] اسمه الكامل نافع وتوفي سنة 116هـ ، أو 117هـ ، أو 118هـ ، أو 119هـ ، أو 120هـ وعاش في مدينة مصر ورتبته الثالثة ونسبه المدني ، القرشي ، العدوي ، وقيل : المغربي الأصل ، وقيل : النيسابوري وقرابته مولى عبد الله بن عمر بن الخطاب ، وأبناؤه : عبد الله ، وعمر ، وأبو بكر • Malik [START] Nafie [END] bin Omar [DESC] His fullname is Nafie. died in 116H, 117H, 118H, 119H or 120H. Lived in Egypt. Ranked third. His lineage is Al-madany, Al-qurashi, Al-adawy, also said to be Al-maghriby and said to be Al-naysabury. Relations: Mawla Abdullah bin Omar bin Al-khattab. His sons: Abdullah, Omar and Abubakr

Table 1 Translated examples of the reranker input

Sanad with 3 targets		
Malik [SEP] Nafie [SEP] bin Omar		
Context-info pair of target narrator Malik	Label	ID
[CLS][START] Malik [END] Nafie...[DESC] His full name is Malik bin Anas...[SEP]	True	664
[CLS][START] Malik [END] Nafie...[DESC] His full name is Malik bin Maghoul...[SEP]	False	153
[CLS][START] Malik [END] Nafie...[DESC] His full name is Malik bin Abdullah...[SEP]	False	940

The ID column displays the generated candidate narrator IDs. The label column indicates the correct narrator

Not all information is provided for all narrators, as some of them may have missing fields. We use the tokens [بداية], [نهاية] ([START], [END]) to indicate the narrator we want to identify. The token [وصف] ([DESC]) separates the sanad and the information.

We feed the context-info pair into pretrained AraBERT, followed by adding a classification layer on top of it. This layer takes the final hidden state of the [CLS] token as input and produces the probability of the candidate and the narrator in the sanad being the same person. Based on the probability scores, the narrator with the highest probability among the candidates is selected as the final prediction.

4.2.1 Training details

The batch size is 32 and the learning rate is 0.0001. We apply early stopping regularization. We use on-plateau learning rate decay. We use Adam optimizer. The classifier was first trained with frozen BERT, and then both were fine-tuned. Huggingface's implementation of AraBERT with default parameter settings was used.² The number of parameters in the reranker is 135,194,882 which take about 540

MB in memory. Training took approximately 5 days on a single GPU (NVIDIA Quadro RTX 6000).

5 Experiments

The experiments use two datasets, the AR-Sanad 280K dataset and the AR-Sanad 280K-v2. Firstly, we performed various experiments on the AR-Sanad 280K dataset to determine the optimal configuration for the model. Then, we trained the model with the best configuration on the AR-Sanad 280K-v2 dataset and compared the performance with that of other methods.

5.1 Generator

This section presents experiments conducted solely on the generator. Previous work [22] used different edge and node operators. We compare our operators to theirs and evaluate the performance. The operators they used are represented by the following equations, which substitute for Eqs. (1) and (2), respectively.

² <https://huggingface.co/>.

Table 2 Experiments with different operators on AR-Sanad 280K dataset

Operators	Accuracy on val
Edge-node	
proj-inter	88.3
trans-inter	88.2
trans-att	90.9
proj-att	88.1

The values in bold indicate the highest scores in each experiment

proj- and -inter are the node and edge operators from Eqs. (3) and (4), respectively. trans- and -att are the node and edge operators from Eqs. (1) and (2), respectively

$$E = R_{\tau} v \quad (3)$$

$$N = W_{\gamma} \Psi(NN_k(E_i), \forall i = 1, \dots, n) \quad (4)$$

where $R_{\tau}^{d \times d}$ is a trainable parameter matrix for edge type τ . NN_k is a 1-layer feedforward neural network. Ψ elementwise mean of a set of vectors. W_{γ} is a trainable transformation matrix for each node type γ .

Tests with different operator combinations: we assess the model's performance by testing it with various operator combinations, as presented in Table 2. Our experimentation covers the use of projection in Eq. (3) and translation in Eq. (1) as edge operators for traversing down the edges, and intersection in Eq. (4) and attention in Eq. (2) as node operators for combining node embeddings.

Tests with different negative samples: at test time, the model selects the most likely narrator among all narrators that share the same appearance form. For training, we experimented with other sets of narrators as negative samples.

- *random30*: Select 30 random narrators as negative samples.
- *sameAppearance*: Select all narrators with the same appearance form
- *sameAppearance+To/From*: Select all the narrators that share the same appearance forms, and also include all the narrators who precede or follow the target narrator in the sanad.

The optimal performance is achieved by using all narrators sharing the same appearance form as negative samples.

Table 3 Training the reranker on the AR-Sanad 280K dataset using various narrator information

Info	Accuracy on val
fullnames	91.8
5 fields	92.8
9 fields	93.5

The values in bold indicate the highest scores in each experiment

Fullnames: narrator's full name only. 5 fields: fullname, byname, death year, death city, living city. 9 fields: 5 fields + rank, lineage, journey cities, relatives

Table 4 Accuracy of the top-performing model on the validation set and test set, trained using the AR-Sanad 280K dataset

Model	Val	Test
Ref. [3]	92.9	83.5
gen	90.9	81
gen+rank	94.6	87.9

The values in bold indicate the highest scores in each experiment

gen is the generator alone. *gen+rank* is the whole model. Compared with model from Ref. [3]

5.2 Reranker

Tests with different fields of narrators associated information: Table 3 displays the performance results of the entire model, which includes the generator and the reranker, trained on various types of candidate information. The generator in these experiments used the original operators presented by Hamilton et al. [22], as indicated in Eqs. (3) and (4). The model was trained on three types of information: (1) full names only, (2) **5 fields**: full name, byname, death year, death city, and living city, and (3) **9 fields**: 5 fields + rank, lineage, visited cities, and relatives. The results indicate that training on information from the 9 fields produces the best performance.

Results of top-performing model: Table 4 displays the results of the top-performing model on both the validation and test sets of AR-Sanad 280K. The table presents two accuracy values: one for the generator alone, denoted as *gen*, and another for the entire model, denoted as *gen+rank*. We compare our results with the results from the model used by Mahmoud et al. [3].

The top-performing model, from here on referred to as ours:

Table 5 Comparing the performance of the models when trained on AR-Sanad 280K dataset versus AR-Sanad 280K-v2

Model	Dataset	Accuracy
MFN baseline		92.5
Frozen AraBERT [3]	AR-Sanad 280K	77.5
Tuned AraBERT [3]	AR-Sanad 280K	83.5
Tuned AraBERT [3]	AR-Sanad 280K-v2	94.1
Ours	AR-Sanad 280K	87.9
Ours	AR-Sanad 280K-v2	95.2

The values in bold indicate the highest scores in each experiment

We test our model, the model introduced by Mahmoud et al. [3] and MFN (Most Frequent Narrator), a baseline that selects the most frequent narrator for each appearance form. The results shown are evaluations on the test set

- Uses translation and attention operators from Eqs. (1) and (2).
- Trained on negative samples that include all narrators with the appearance form.
- Trained on nine fields of information.

5.3 Comparative results on AR-Sanad 280K-v2

We train our model and the model from Mahmoud et al. [3] on the AR-Sanad 280K-v2 and compare the performance in Table 5. In Ref. [3], they test AraBERT in two settings: a frozen setting where only the classification layer is trained and a tuned setting where the whole model is trained. We report the

results from both settings. Inspired by the MFS baseline (Most Frequent Sense) in Word Sense Disambiguation, we also test the MFN baseline (Most Frequent Narrator). The MFN baseline is a strong baseline that selects the most frequent narrator for each appearance form. Data about appearance forms and their most frequent narrator are collected from (alifta.net). The results show 10% improvement by using the updated version of the dataset and our system outperforms the previous system by using only 30% more parameters.

5.4 Experiments using ChatGPT

We sampled 10 sanads from the test set and asked ChatGPT to identify the narrators. There were 43 narrators and only 14 were identified correctly. In Fig. 10, we show responses from ChatGPT when asked to identify narrators in different sanads. The last sanad contains 6 narrators with only their first names mentioned. ChatGPT only gives the first names or states that there is not enough information. We tested 10 times with 3 different prompts for this sanad. Another issue is that ChatGPT sometimes gives the lineage of the narrator (crossed out text) which makes it hard to link the name with a narrator entity.

6 Discussion and further analysis

In Table 6, we report the precision, recall, and additional evaluation metrics outlined by Mahmoud et al. [3]. Given the significance of accurately identifying ALL narrators in a sanad, the Sanad Error Rate (SER) is a crucial metric to consider, which is defined as the percentage of sanads that

Fig. 10 Responses from ChatGPT when asked to identify narrators in different sanads. We use the prompt " اذكر اسم الراوي كاملا لكل راوي في هذا "السند: السند" which translates to "Please write the full name of each narrator in the sand: [sanad]"

آدم [فاصل] شعبة [فاصل] الأعمش [فاصل] أبي وائل [فاصل] حذيفة • آدم بن أبي إياس الأنصاري ✓ • شعبة بن الحجاج اللبثي ✓ • الأعمش بن عبد الله الأنصاري ✗ • أبي وائل شعبة بن عبد الله الأنصاري ✗ • حذيفة بن اليمان الأنصاري ✓	مسدد [فاصل] حماد [فاصل] هشام [فاصل] أبيه [فاصل] عائشة • مسدد بن معد بن تميم ✗ • حماد بن سلمة بن كهيل ✗ • هشام بن عمار بن عمير ✗ • أبيه بن السفر بن حبيب ✗ • عائشة بنت زياد بن زرارة ✗
آدم [فاصل] شعبة [فاصل] الحكم [فاصل] إبراهيم [فاصل] الأسود [فاصل] عائشة ✗	سليمان بن حرب [فاصل] شعبة [فاصل] قتادة [فاصل] أنس • سليمان بن حرب • شعبة بن عياش الجرشي ✗ • قتادة بن دعامة ✓ • أنس بن مالك ✓

Table 6 Precision, recall, sanad error rate (SER), and top k accuracy on the test set

Model	Ours	Mahmoud	MFN
Precision	82.7	–	83.4
Recall	84.7	–	82.7
F1	82.8	78.8	81.9
SER	23.2	60.6	33.6
Accuracy	95.2	83.5	92.5
Top-3 accuracy	97.1	95.3	97.9
Top-5 accuracy	97.2	97.1	98.2

The values in bold indicate the highest scores in each experiment

Evaluation of our model trained on AR-Sanad 280K-v2 dataset, the model from Mahmoud et al. [3] trained on the AR-Sanad 280K dataset and the MFN baseline

Table 7 Percentage of sanads with n misidentified narrators

# Misses	Percentage (%)
0 narrators	76.8
1 narrator	20.2
2 narrators	2.7
>2 narrators	0.3

Table 8 Effect of sanad length on accuracy

Sanad length	Accuracy (%)
3 narrators	95.7
4 narrators	95.3
5 narrators	95.2
6 narrators	95.5
7 narrators	95.1
>7 narrators	94.8

• نصر بن علي [فاصل] خالد [فاصل] سعيد [فاصل] قتادة [فاصل] أنس

- Nasr bin Ali [SEP] Khalid [SEP] Saied [SEP] Katada [SEP] Anas
- Narrator IDs: [955, 2382, 365, 369, 8]

Fig. 11 A sanad and its corresponding list of narrator IDs from the test set. The predictions of the models on this sanad are presented in Table 9

contain at least one incorrectly classified narrator. Notably, our model demonstrates a significantly lower error rate compared to both Mahmoud et al. [3] and the MFN baseline. Table 7 provides data regarding the percentage of sanads that contain n misidentified narrators.

In contrast to the findings reported by Mahmoud et al. [3], which indicated an increase in accuracy as the length of the sanad increased, our observations from Table 8 demonstrate that the performance of our model is not significantly impacted by the length of the sanad. Table 9 compares the predictions of MFN, the model from Mahmoud et al. [3], and our model on one of the sanads from the test set shown in Fig. 11. In this particular sanad, all narrators except the first one are referred to using only their first name, making it a challenging case. However, our model correctly identified all of them.

7 Conclusion

In conclusion, our paper presents a novel model designed to leverage the graph structure of the sanad for narrator identification, resulting in a notable performance enhancement. By constructing a graph representing narrators' interactions and appearance forms, we facilitated the training of our model. We also update the AR-Sanad 280K dataset for better representation of real hadiths. Our experiments on the updated AR-Sanad 280K dataset demonstrated significant improvements, achieving an accuracy of 95.2% and a sanad error rate (SER) of 23.2% on the test set. In comparison, the model proposed by Mahmoud et al. [3] attained an accuracy of 83.5% and a SER of 60.6%. These findings underscore the efficacy of our approach in enhancing the accuracy and reliability of narrator disambiguation tasks within the context of hadith studies. Moving forward, our model opens avenues for further research in this domain and offers valuable insights into improving the robustness of narrator identification systems.

To further improve performance, a possible future direction is to make use of the meaning of special appearance forms. These forms are used when a narrator refers to other narrators by their relation, such as أبيه which means "his father." Although our model can disambiguate those special

Table 9 Comparing predictions of three models on the sanad depicted in Fig. 11

Narrator	MFN	Ref. [3]	Ours
Narr: 955	Nasr bin Ali bin Nasr..	Nasr bin Ali bin Nasr..	Nasr bin Ali bin Nasr..
Narr: 2382	Khalid bin Mahran	Khalid bin Quays bin Rabah	Khalid bin Al-harith..
Narr: 365	Saied bin Mansour bin Shu'ba	Saied bin Abi Orouba..	Saied bin Abi Orouba..
Narr: 369	Katada bin Da'ama..	Katada bin Da'ama..	Katada bin Da'ama..
Narr: 8	Anas bin Malik bin Al-Nadr..	Anas bin Malik bin Al-Nadr..	Anas bin Malik bin Al-Nadr..

The bold text indicates an incorrect prediction

forms, it doesn't take full advantage of their meaning as it deals with them like other appearance forms. If we can extract those relations from narrators' information and add them to the graph, this will enable us to make use of their meaning.

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Data availability Data will be provided at <https://github.com/somaia02/Narrator-Disambiguation>.

Declarations

Conflict of interest The authors have no conflict of interest to declare.

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